

Radioastronomy

Fortgeschrittenenpraktikum

HS1 Radioastronomy

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1 Introduction

The experiment HS1-Radioastronomy is there to give an overview for operating a radio telescope and to introduce the prerequisites such as characterising and calibrating the telescope before carrying out the experiments. The main experiment to be conducted is the observation of neutral hydrogen in the Milky Way to determine the rotational velocity in order to derive the mass distribution.

2 Theoretical Background

2.1 Radioastronomy

Radio astronomy is the study of electromagnetic radiation in the radio wavelength range, typically from a few MHz to several hundred GHz. Compared to optical astronomy, radio waves are much less affected by interstellar dust and gas, which allows astronomers to observe cold and dense regions of space that are often hidden in the optical regime. Radio telescopes usually consist of large parabolic antennas that collect incoming radiation and direct it to a receiver. The receiver amplifies the weak signal and converts it into an electrical voltage, from which the intensity, spectral distribution, and temperature of the observed source can be determined.

A central quantity in radio astronomy is the brightness temperature. In the Rayleigh-Jeans regime, which is valid for most radio astronomical observations, the spectral intensity is approximately

$$I_\nu \approx \frac{2k_B T_B \nu^2}{c^2} = \frac{2k_B T_B}{\lambda^2},$$

where I_ν is the spectral intensity, k_B is the Boltzmann constant, T_B is the brightness temperature, ν is the frequency, and c is the speed of light. This relation shows that the observed radio signal is directly linked to the temperature of the emitting medium. This makes radio astronomy a powerful method for studying cold molecular clouds, ionized gas regions, and neutral hydrogen structures in the Milky Way and beyond.

One of the most important spectral lines in radio astronomy is the 21 cm line emitted by neutral hydrogen.

2.2 The 21 cm Hydrogen Line

The hydrogen atom consists of a proton and an electron. In the ground state, the proton and electron spins can be oriented either parallel or antiparallel to each other. This gives rise to a hyperfine splitting of the ground state: the antiparallel configuration has a slightly lower energy than the parallel configuration. A transition from the higher-energy state to the lower-energy state leads to the emission of a photon with a frequency of approximately

$$\nu \approx 1420.4 \text{ MHz},$$

which corresponds to a wavelength of

$$\lambda \approx 21.1 \text{ cm}.$$

This transition is therefore called the 21 cm line.

This line is of great importance in astronomy because hydrogen is the most abundant element in the interstellar medium. Neutral hydrogen emits strongly at 21 cm, allowing astronomers to map the distribution of gas in galaxies. The 21 cm line is especially useful because it is only weakly absorbed by interstellar dust, so it can reveal regions that are difficult or impossible to observe in the optical range.

The 21 cm line also enables the measurement of Doppler shifts. If the hydrogen gas is moving relative to the observer, the frequency of the emission is shifted. From this shift, the radial velocity of the gas can be determined. This makes it possible to study galactic rotation, gas dynamics, and the mass distribution of galaxies. For this reason, the 21 cm line is one of the most important tools in modern radio astronomy and is fundamental for studying neutral hydrogen in the universe.

3 Observation and Evaluation

3.1 The KRT3 telescope

3.2 Weather Conditions

The experiment was carried out on 07.09.2026, starting at around 12:35. The weather was stable throughout the day with partial cloud coverage and temperatures around 19°.

3.3 Calibration

As the receiver receives radiation not only from the source T_S but also from the environment T_h . In addition to those two external sources of radiation, the antenna also picks up on the radiation it's own system emits T_{sys} . This means the measured power of the receiver, represented by the antenna temperature, can be broken down the following way:

$$T_A = T_S + T_h + T_{sys} \quad (1)$$

It is difficult to separate the pure system temperature and the background radiation, so these temperatures are usually combined into an effective system temperature:

$$T'_{sys} = T_{sys} + T_h \quad (2)$$

The effective system temperature now includes all unwanted contributions. This means that to know the contribution of a source, one must know the effective system temperature. Determining the effective system temperature is calibrating the telescope by doing a 'On-Off' measurement through measuring a source with a known flux density and then measuring a piece of empty sky. The KRT3 however, can only measure the sun as a source with high enough flux density but the sun is not a constant source which is why a noise generator needs to be used.

The noise generator injects a signal directly into the antenna horn. The calibration temperature $T_{cal} = 170\text{ K}$ is the expected antenna temperature that is observed additionally when the noise generator is switched on.

When assuming the sun to be a point-like source ?? can be used in the form of:

$$T_b \Omega_s = T_A \Omega_A \quad (3)$$

with T_b being the brightness temperature, Ω_s being the solid angle of the sun, T_A being the antenna temperature and Ω_A being the solid angle of the antenna main lobe.

The goal is measuring the antenna temperature of the sun and converting it into a flux density. For this we need the following identities: the antenna theorem $A_e \Omega_A = \lambda^2$, the Rayleigh-Jeans law $B_\nu = 2kT_b/\lambda^2$, the flux density $S_\nu = B_\nu \Omega_s$ and the aperture efficiency $\eta = A_e/A_g = 0.5$ with A_g being the geometric antenna surface assumed to be a circle.

$$T_b \Omega_s = T_A \Omega_A \quad (4)$$

$$\Omega_A = \frac{\lambda^2}{A_e}, \Omega_s = \frac{S_\nu}{B_\nu}, B_\nu = \frac{2k_B T_b}{\lambda^2} \quad (5)$$

$$S_\nu = \frac{T_A \cdot 2k_B}{A_e} = f(T_A, A_e) \quad (6)$$

$$A_e = \pi r^2 \cdot \eta = 1,5^2 \pi \cdot 0,5 = \pi 1,125 \text{ m}^2 \quad (7)$$

$$\Rightarrow T_A = \frac{S_\nu A_e}{2k_B} = \frac{S_\nu \pi 1,125 \text{ m}^2}{2k_B} \quad (8)$$

3.3.1 Determination of the system temperature

3.3.2 Background Radiation

3.4 Determination of the angular resolution of the KRT3

The expected half-width is $q = 1.02\lambda/D = \dots$ with the wavelength $\lambda = \dots$ and the antenna diameter $D = 3\text{ m}$.

3.5 Determination of the Sensitivity of the instrument

To determine the theoretical sensitivity of the instrument, the weakest detectable flux density ΔS_ν for a given system temperature T'_{sys} can be given by the radiometer equation:

$$\Delta S_\nu = \frac{2k}{A_e} \frac{C_s T'_{sys}}{\sqrt{\Delta\nu\tau}} \quad (9)$$

where $C_s \approx 2$ is a dimensionless instrumental constant and τ the integration time. With Gleichung 9 the limiting sensitivity in a single channel with $\Delta\nu = 61$ kHz and an integration time of $\tau = 1$ s can be calculated to $\Delta S_\nu = \dots$

The measurement is carried out with the radio source Vir A with an elevation of $\dots^\circ$ at the time of measurement. The necessary integration time is calculated to be $\tau = \dots$, taking the flux density at $\nu = 1420,4$ MHz to be $S_\nu = 211$ Jy $\pm$ 11 Jy after Baars et al. (1977).

3.6 Observation of neutral hydrogen in the Milky Way

The preamplifier has a frequency range of 1398.00-1460.56 MHz divided into 1024 channels. The frequency resolution of a single channel (the bandwidth) is calculated ... As we observe the H1 line in a single channel, the velocity resolution is ... If we assume a radial velocity of 0 km/s

4 Results

4.0.1 Determination of the system temperature

4.0.2 Background Radiation

4.1 Determination of the angular resolution of the KRT3

4.2 Determination of the Sensitivity of the instrument

4.3 Observation of neutral hydrogen in the Milky Way

5 Discussion

6 Summary

Literatur

Abbildungsverzeichnis

Labbook

Fortgeschrittenenpraktikum HS1 Radioastronomy

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07. - 09. September 2026

Wetter

Datum: 07/09/26

Uhrzeit: 12:35 Uhr



- Teilweise bedeckter Himmel
- Kein Niederschlag

5.1.1 Calibration (Systemtemperatur)

USB 1-5, 9-13

Datendatei-Struktur im Programm (RadioUniversePro):

rep	dev	sb	on_tbi	off_tbi	cal_tbi	flux	SEFD
1	bbc01	u	12195,30	1001,40	2634,20	00,00	00,00

$$T_b \cdot \Omega_S = T_A \Omega_A \quad (10)$$

$$\Omega_A = \frac{\lambda^2}{A_e}, \quad \Omega_S = \frac{S_\nu}{B_\nu}, \quad B_\nu = \frac{2k_B T_b}{\lambda^2} \quad (11)$$

$$S_\nu = \frac{T_A \cdot 2k_B}{A_e} = f(T_A, A_e) \quad (12)$$

$$A_e = \pi r^2 \cdot \eta = 1,5^2 \pi \cdot 0,5 = 1,125 \pi \text{ m}^2 \quad (13)$$

$$T_A = \frac{S_\nu A_e}{2k_B} \quad (14)$$

$$T_{\text{sys, left}} = 98.76 \pm 22.54 \text{ K}$$

$$T_{\text{sys, right}} = 107.80 \pm 31.65 \text{ K}$$

$$T_{\text{sun, left}} = 1237.79 \pm 89.62 \text{ K}$$

$$T_{\text{sun, right}} = 1302.59 \pm 119.16 \text{ K}$$

Mit S_ν der Sonne (aus SWC):

- 76 sfu (solar flux units): $76 \cdot 10^{-11} \text{ W m}^{-2} \text{ Hz}^{-1} = 76 \cdot 10000 \text{ Jy} = 760000 \text{ Jy}$
- $\Rightarrow T_A = 972, 753 \text{ K}$
- Störung wegen RFI

5.2 Angular Resolution

USB 1-5, 9-13

Position der Sonne berechnet durch die Steuerungssoftware:

Source	RA	DEC	AZ	EL	HA	Type	Note
Sun	110410.1	+055810.9	+1684733.7	+42 00 48.8	-003253.9	Star	East

- Im Ordner `cscan` befindet sich der Screenshot.
- Im Ordner `FITS` befinden sich die Daten (lon vs. lat).

5.3 Sensitivity of the Instrument

Antennentemperatur von VirA (BBC 3,11 zeigt geringsten RFI-Einfluss).

Benötigte Zeit für das Mapping:

- 28:35.4s (Software)
- 29:27.8s (Eigener Timer)

Hinweis: Screenshot auf Ubuntu und nicht Windows! Files im Ordner `maps` und in `FITS` als `imageproJ01 VirA 01`.

TODO: Theoretical Sensitivity

$$\Delta S_\nu = \frac{2k}{A_e} \frac{C_S T'_{\text{sys}}}{\sqrt{\Delta\nu\tau}} \quad (\text{spektrale Flussdichte}) \quad (15)$$

Mit $C_S \approx 2$, $T'_{\text{sys, left}} = 98.76 \pm 22.54 \text{ K}$, $T'_{\text{sys, right}} = 107.80 \pm 31.65 \text{ K}$, $A_e = 1.125\pi$, $\tau = 1\text{s}$, $\Delta\nu = 61\text{kHz}$:

$$\Delta S_{\nu, \text{left}} = 6,248 \cdot 10^{-24} \frac{\text{W}}{\text{m}^2 \text{Hz}} = 624,92 \text{ Jy} \quad (16)$$

$$\Delta S_{\nu, \text{right}} = 6,82 \cdot 10^{-24} \frac{\text{W}}{\text{m}^2 \text{Hz}} = 682,02 \text{ Jy} \quad (17)$$

Benötigte Integrationszeit τ zum Detektieren der ausgewählten Quelle mit dem KRT 3 mit einem Signal-Rausch-Verhältnis $S/N = 10$ in jedem Kanal:

- Virgo A: Flussdichte = $211 \pm 11 \text{ Jy}$, Radiogalaxie
- Für 5 Kanäle jede Seite ($\Delta\nu = 5 \cdot 61\text{kHz} = 305\text{kHz}$)
- $S/N = 10$
- Formel: $\tau = \left(\frac{2k \cdot C_S \cdot T'_{\text{sys}}}{\Delta S_\nu \cdot A_e} \right)^2 \cdot \frac{\Delta\nu}{S/N}$
- $\tau_l = 4,12\text{s}$
- $\tau_r = 5,22\text{s}$

Determination of the Uncertainty of Radiant Flux Measurements

Preliminary Work

$$S_\nu = \frac{P}{4\pi d^2 \sqrt{2\pi\sigma}} = 1263,16 \text{ Jy} \quad (18)$$

Für Mobile Radio on the Moon (übertragen mit 2W, 900MHz, FWHM: 80kHz, $\sigma = 34\text{kHz}$, empfangen auf der Erde).

$$S_\nu \propto \nu^\alpha, \quad \alpha = -0,7 \quad (19)$$

Bei $\nu = 1420.4 \text{ MHz}$: $S_{\nu, \text{CasA}} = 1420 \pm 70 \text{ Jy}$.

$$c = \frac{1420,4^{-0,7} \text{ MHz}}{1420 \text{ Jy}} \quad (20)$$

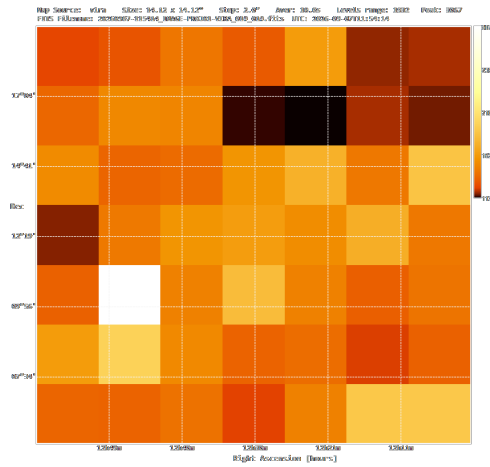
$$S_{\nu, \text{CasA}} = \frac{900^{-0,7}}{c} \text{ Jy} = 1954,37 \text{ Jy} \quad (21)$$

$$T_A = \frac{S_\nu A_e}{2k_B} = 1,82 \text{ K} \quad (\text{für } \nu = 1420.4 \text{ MHz}) \quad (22)$$

TODO: Fehlerfortpflanzung

Evaluation

BBC 3,11



```
center=-1.8189401 0.015094938
wcs
1 pixel = 1
reg sum error area surf_bri surf_err
-----
1 50954 225.73 35 1455.83 6.44943
reg sum npix mean median min max var stddev rms
-----
1 50954 35 1455.83 1432 1135 1896 45031.3 212.206 1471.21
```

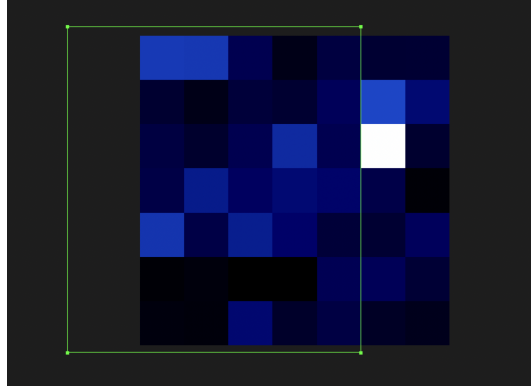
$$\text{mean} = 1455,83 \quad (23)$$

$$\text{std} = 212,206 \quad (24)$$

$$\sigma[K] = \frac{\text{std}}{\text{mean}} \cdot T'_{\text{sys}} = 15,71 \text{ K} \quad (25)$$

$$\text{mean}_{\text{scaled}}[K] = T_A(\text{OFF}) = 156938,474 \text{ K} \quad (26)$$

$$T_A(\text{ON}) = 3067 \cdot 107,8\text{K} = 330622,6 \text{ K} \quad (27)$$



Antennentemperatur der Quelle:

$$T_s = \frac{T_A(\text{ON}) - T_A(\text{OFF}) + T'_{\text{sys}}}{T'_{\text{sys}}} = 1612,17 \text{ K} \quad (28)$$

- Rauschen ist viel größer als erwartet.
- Vergleich mit theoretischer Empfindlichkeit (600 Jy, ca. 1 K).
- Bei uns jetzt 128 Kanäle, 10s Integrationszeit → eigentlich sollte es noch besser sein ($\ll 1$ K).
- Bei uns jedoch viel Einfluss von RFI → ~ 15 K Std./Rauschen.
- Empfänger-/Verstärkersensibilität ist ein Problem; thermisches Rauschen begrenzt nicht, aber besser als 1 K kann es eigentlich nicht werden.

Gesucht: S/N (25 Karten für $S/N = 10$).

5.1.2 Background Radiation

Mond

Elevation: $30^\circ 52' 32.0''$ (in Grad, Minuten, Sekunden)

Für BBC 1-5, 9-13 (wie zuvor für die Sonne, auch bei schlechter Qualität).

$$\text{Span} = 2(\text{ELV} - 10^\circ) = 2(30^\circ - 10^\circ) = 40^\circ \quad (29)$$

Da der Azimut bei $268^\circ 03' 32.7''$ liegt, wählen wir einen etwas kleineren Span: $\text{Span} = 38^\circ$.

Zeitpunkt: ca. 14:56 Uhr.

Daten in FITS: LON-MOON und LAT-MOON + Screenshot in `cscan`: CSCAN-MOON.

Virgo A

Zeitpunkt: 15:28 Uhr

Elevation: $47^\circ 56' 18''$, Azimut: 196° , Span = 74.

Observation of Neutral Hydrogen in the Milky Way

Frequency Setting of the Receiver

Preliminary Work

- Was ist die Frequenzauflösung $\delta\nu$ (Bandbreite eines einzelnen Kanals) und wie lautet die Geschwindigkeitsauflösung δv bei der Beobachtung der H I-Linie?
- In welchem Kanal würden Sie die H I-Linie bei $\nu_0 = 1420.4$ MHz erwarten, unter Annahme einer Radialgeschwindigkeit von 0 km s^{-1} ?

- Sie wollen das Radiokontinuum einer Quelle messen. Welche Frequenzen sind innerhalb der Bandbreite erlaubt, damit die Messung nicht durch die 21cm-Linie kontaminiert wird? (Maximale Radialgeschwindigkeit des H I: $\pm 200 \text{ km s}^{-1}$).
- Welche Frequenzbänder sollten in BBC Tools ausgewählt werden, wenn man einmal die H I-Linie und einmal das Kontinuum beobachten möchte?

Measurement

$$R = R_0 \cdot \sin(l), \quad R_0 = 8,5 \text{ kpc} \quad (30)$$

l [deg]	R [kpc]	$v_{\max}$ [km/s]	v_R [km/s]
10	$1,48 \pm 4,09$	–	–
20	$2,91 \pm 4,0$	$70,70 \pm 14,50$	...

- $v_{\max}$ grafisch bestimmt (Schnittwert mit der orangen σ -Linie).
- $v_R = v_{\max} + \omega_0 R_0 \cdot \sin(l)$ mit $\omega_0 R_0 = 220 \text{ km/s}$.
- $v_{\max, \text{err}}$ grafisch über blauen Bereich bestimmt (Schnittpunkt min – Schnittpunkt max / 2).
- $\Delta l = \pm 2^\circ$.
- Fehlerfortpflanzung für R :

$$\Delta R = \left| \frac{\partial R}{\partial l} \right| \cdot \Delta l = |R_0 \cdot \cos(l)| \cdot \Delta l \quad (31)$$

$$\sigma_R = \sqrt{|R_0 \cos(l)| \Delta l} \quad (32)$$

- Eingeschlossene Masse:

$$M(R) = 0,234 \cdot 10^{10} \left(\frac{v(R)}{100 \text{ km/s}} \right)^2 \left(\frac{R}{\text{kpc}} \right) M_\odot, \quad 1M_\odot = 1,98 \cdot 10^{30} \text{ kg} \quad (33)$$

Distribution of Dark Matter

$$M_{\text{DM}}(R_0) = 66411575447.22 \pm 3593349279.00 M_\odot \quad (34)$$